MANGU 2025 KCSE PREDICTION

CONFIDENTIAL EXAM

233/3

CHEMISTRY

PAPER 3 (PRACTICAL)

TIME: 2¼ HOURS

NAME……………………………………………………………………………

SCHOOL………………………………………………… **SIGN**………………………

INDEX NO……………………………………………… **ADM NO**……………………

(Kenya Certificate of Secondary Education)

Instructions:

- *Answer **ALL** questions in the spaces provided.*
- *You are **NOT** allowed to start working with the apparatus for the first 15 minutes of the 2 ¼ hours allowed for this paper. This time will enable you read through the question paper and make sure you have all the chemicals and apparatus required.*
- *Mathematical tables and electronic calculators may be used*
- *All working must be clearly shown where necessary.*

FOR EXAMINER'S USE ONLY.

Question	Maximum score	Candidate's score
1	20	
2	12	
3	8	
Total score	**40**	

1. You are provided with:

- Solution A of Potassium manganate (VII).
- 0.05M solution B of oxalic acid.
- Solution C containing 4.9g of ammonium iron (II) Sulphate, $(NH_4)_2 SO_4.FeSO_4.6H_2O$, in 250cm^3 of water.
- You are required to:

i) Determine the rate of reaction between oxalic acid and Potassium manganate (VII) solutions.

ii) Standardize the solution A.

PROCEDURE I:

(i) Fill the burette with solution A.

(ii) Place 1 cm^3 of solution A from the burette into each of the five (5) test-tubes in a test tube rack.

(iii)Using a clean measuring cylinder, place 19 cm^3 of solution B into a boiling tube.

(iv)Place the thermometer into solution B and heat gently until it attains a temperature of 40 ^{0}C.

(v) Add the first portion of solution A **immediately and at the same time start a stop watch**.

(vi) Record the time taken for solution A to be decolorized in table I below.

(vii) Repeat the procedure (i) to (v) at temperatures of 50^{o}C,60^{o}C,70^{o}C and 80^{o}C to complete the table.

Table I

Temperature of solution B (^{o}C)	40	50	60	70	80
Time taken for decolorization (seconds)					
Rate ($^I/_t$) s^{-1} X 1000					

(4 marks)

i) Plot a graph of ($^{1}/_{t}$ X 1000) against temperature (X-axis)

(3 marks)

ii) From the graph determine the time taken for the mixture to decolourise at 65^{o}C **(2 marks)**

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iii) How does the rate of reaction between oxalic acid solution B and Potassium manganate (VII) solution A vary with temperature? **(1 mark)**

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PROCEDURE II

-Refill the burette with solution A.

-Pipette 25cm^3 of solution C into a conical flask and titrate the solution A against solution C until a permanent pink colour just appears.

-Record your results in table II below and repeat the procedure to fill the table.

Table II	1	2	3
Final burette reading (cm^3)			
Initial burette reading (cm^3)			
Volume of A used (cm^3)			

(4 marks)

i) Determine the average volume of A used. **(1 mark)**

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ii) Calculate the concentration of solution C in moles per litre (Fe = 56, S = 32, O = 16, N =1 4, H = 1) **(1 mark)**

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iii) Find the number of moles of solution C used. **(1 mark)**

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iv) Given the ionic equation for the reaction is

$$5Fe^{2+}_{(aq)} + MnO_4^-{}_{(aq)} + 8H^+_{(aq)} \longrightarrow 5Fe^{3+}_{(aq)} + Mn^{2+}_{(aq)} + 4H_2O_{(l)};$$

Find the number of moles of solution A used. **(1 mark)**

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(v) Determine the concentration of the Potassium manganate (VII), solution A in moles per litre.

(2 marks)

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2. You are provided with solid Q. Carry out the tests below and record your observations and inferences in the table below.

i) Place half a Spatula full of solid Q in a clean dry test-tube and heat gently then Strongly.

Test the gas produced using moist red and blue litmus papers

Observations	**Inferences**
(1 mark)	**(1 mark)**

ii) Place the remaining solid Q in a boiling tube, add about $5cm^3$ of distilled water and shake well. Divide the resulting mixture into four portions for the tests below.

Observations	**Inferences**

(1 mark)	(1 mark)

a). To the first portion add Sodium hydroxide solution dropwise until in excess.

Observations	**Inferences**
(1 mark)	**(1 mark)**

b). To the second portion add 2-3 drops of dilute Sulphuric (VI) acid

Observations	**Inferences**
(1 mark)	**(1 mark)**

c). To the third portion add aqueous ammonia dropwise until in excess

Observations	**Inferences**
(1 mark)	**(1 mark)**

d). To the fourth portion add 2-3 drops acidified barium nitrate solution

Observations	Inferences
(1 mark)	**(1 mark)**

3. You are provided with solid L. carry out the tests below on L and record the observations and inferences in the spaces provided.

(a) Place **half** of solid L in a boiling tube and add about $5cm^3$ of distilled water

(i)

Observations	Inferences
(1 mark)	**(1 mark)**

Divide the solution into two portions and carry out the tests below.

(ii). To the first portion add 2-3 drops of acidified potassium manganate (VII).

Observations	Inferences
(1 mark)	**(1 mark)**

(iii). To the second portion add Sodium carbonate provided.

Observations	Inferences

(1 mark)	**(1 mark)**

(b). Place the remaining solid L in metallic spatula and ignite it.

Observations	**Inferences**
(1 mark)	**(1 mark)**

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